



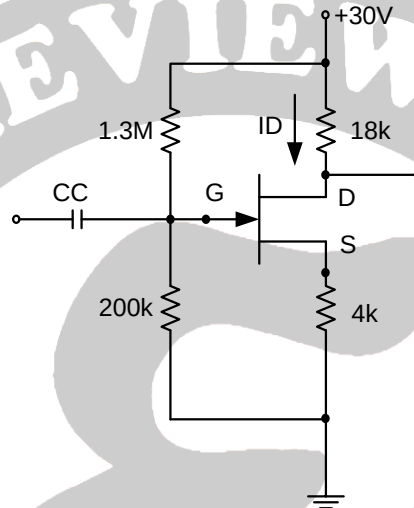
Multiple Choice:

- A JFET parameter that describes how effective the gate-source voltage is in controlling the drain current is called its
 - gamma, γ .
 - Beta, β .
 - transconductance, g_m .
 - None of the above.
- The normal operation of JFET is
 - constant voltage region
 - constant current region
 - both constant voltage and constant current regions
 - either constant voltage or constant current region
- The depletion-mode MOSFET
 - can operate with only positive gate voltages
 - can operate with only negative gate voltages
 - cannot operate in the ohmic region
 - can operate with positive as well as negative gate voltages
- To avoid damaging MOSFETs during handling,
 - always wear a grounded wrist strap.
 - never apply an input signal when the dc power supply is OFF.
 - never insert or remove them from a circuit when the power is ON.
 - All of the above
- In a p-channel FET, the charge carriers are _____.
 - holes
 - electrons
 - both holes and electrons
 - ions
- The pinch-off voltage of a FET is about _____.
 - 15 V
 - 25 V
 - 0.6 V
 - 5 V

7. A depletion-type MOSFET is a

- normally OFF device.
- normally ON device.
- current-controlled device.
- None of the above.

8. In the circuit shown below, $I_{DSS} = 4\text{mA}$, $V_p = 4\text{V}$. Find the quiescent value of I_D of the FET.



- 0.25 mA
- 0.35 mA
- 0.15 mA
- 0.02 mA

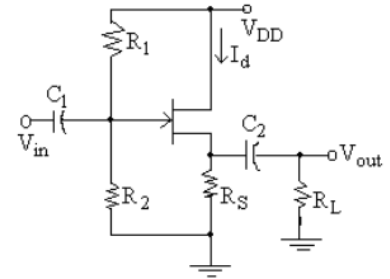
9. When drain voltage equals the pinch-off voltage, then drain current _____ with the increase in drain voltage.

- decreases
- increases
- remains constant
- None of the above

10. A JET operates in the current-source region when

- $V_{GS} > V_p$.
- $V_{GS} < V_p$.
- $V_{GS} = 0\text{V}$.
- $V_{GS} = 0\text{V}$.

11. The FET circuit given below in the figure, has $R_1 = 3.5\text{M}\Omega$, $R_2 = 1.5\text{M}\Omega$, $R_S = 2\text{k}\Omega$, $R_L = 20\text{k}\Omega$ and $g_m = 2.5\text{mS}$. Find its input impedance.



- 1.05 M Ω
- 1.50 M Ω
- 1.01 M Ω
- 1.10 M Ω

12. If the across-sectional area of the channel in n-channel FET increases, the drain current _____.

- remains the same
- is decreased
- is increased
- None of the above

13. A JFET is a

- unipolar device.
- voltage-controlled device.
- current controlled device.
- Both A and B.

14. Which of the following types of bias produces the most unstable Q point in a JFET amplifier?

- gate bias.
- current-source bias.
- voltages divider bias.
- self-bias.

15. Which JFET amplifier is also known as a source follower?

- The common-source amplifier.
- The common-gate amplifier.
- The common-channel amplifier.
- The common-drain amplifier.

16. The formula for drain resistance of a FET is _____.

- $\frac{\Delta V_{DS}}{\Delta I_D}$ at constant V_{GS}
- $\frac{\Delta V_{GS}}{\Delta I_D}$ at constant V_{DS}
- $\frac{\Delta I_D}{\Delta V_{GS}}$ at constant V_{DS}



- D. $\frac{\Delta I_D}{\Delta V_{DS}}$ at constant V_{GS}
17. The pinch-off voltage in a FET is analogous to _____ voltage in a vacuum tube.
- A. anode voltage
B. cathode voltage
C. grid cut off voltage
D. None of the above
18. A FET is characterized by _____ gain.
- A. voltage
B. current
C. resistance
D. None of the above
19. In a JFET amplifier with self-bias,
- A. $V_B = 0$ V.
B. $V_S = I_D R_S$.
C. $V_{GS} = -I_D R_S$.
D. All of the above.
20. In which JFET amplifier are the ac input and output voltages 180° out of phase?
- A. The common-gate amplifier.
B. The common-source amplifier.
C. The common-drain amplifier.
D. The source follower.
21. Given $k = 0.4 \times 10^{-3} \text{ A/V}^2$ and $I_{D(on)} = 3 \text{ mA}$ with $V_{GS(on)} = 4 \text{ V}$, determine V_T .
- A. 1.261 V C. 1.361 V
B. 1.461 V D. 1.561 V
22. The value of drain to source voltage, V_{DS} , at which the drain current, I_D , levels off is called the
- A. cutoff voltage, $V_{GS(off)}$
B. pinch-off voltage, V_p .
C. breakdown voltage, V_{BR}
D. threshold voltage $V_{GS(th)}$
23. For a depletion-type MOSFET with zero bias, the drain current, I_D , equals.
- A. zero.
B. I_{DSS} .
C. $\frac{1}{2} I_{DSS}$.
D. It cannot be determined.
24. The input impedance of a FET is _____ that of an ordinary transistor.
- A. equal to
B. more than
C. less than
D. None of the above
25. The input impedance of a MOSFET is
- A. higher than that of a JET.
B. lower than that of a JFET.
C. no different than that of a JFET.
D. approximately zero ohms.
26. An enhancement-type MOSFET is a
- A. normally OFF device.
B. normally ON device.
C. low input impedance device.
D. current-controlled device.
27. The transconductance of a FET ranges from _____.
- A. 0.5 to 30 m A/V
B. 100 to 500 m A/V
C. 500 TO 10000 m A/V
D. Above 10000 m A/V
28. The drain and source leads may be interchanged when using a (n)
- A. asymmetrical JFET.
B. symmetrical JFET.
C. D-type MOSFET.
D. None of the above.
29. If properly biased JFET acts as
- A. Current controlled current source
B. Voltage controlled voltage source
C. Voltage controlled current source
D. Current controlled voltage source
30. To get a negative gate-source voltage in a self-biased JFET circuit, you must use a
- A. voltage divider
- B. source resistor
C. ground
D. negative gate supply voltage
31. A FET connected in _____ configuration acts as an emitter follower.
- A. common drain
B. common gate
C. common source
D. none of the above
32. The best location for setting a Q-point on dc load line of an FET Amplifier is at
- A. Saturation point
B. Cutoff point
C. Mid- point
D. None of these
33. Which of the following JFET amplifiers has the lowest input impedance?
- A. The common-gate amplifier.
B. The common-source amplifier.
C. The common-drain amplifier.
D. The source follower.
34. If the reverse bias on the gate of a FET is increased, then width of conducting channel _____.
- A. remains the same
B. is decreased
C. is increased
D. None of the above
35. Given $I_D = 14 \text{ mA}$ and $V_{GS} = 1 \text{ V}$, determine V_p if $I_{DSS} = 9.5 \text{ mA}$ for a depletion-type MOSFET.
- A. -4.67 V C. -3.67 V
B. -2.67 V D. -1.67 V
36. A FET has three terminals, namely _____.
- A. cathode, anode, grid
B. emitter, base, collector
C. source, gate, drain
D. None of the above



37. If the gate of a FET is made less negative, the width of the conducting channel_____.

- A. remains the same
- B. is decreased
- C. is increased
- D. None of the above

38. A MOSFET can be operated with _____.

- A. negative gate voltage only
- B. positive gate voltage and negative gate voltage
- C. with both positive and negative gate voltage
- D. None of the above

39. For an enhancement-type MOSFET, the threshold voltage, $V_{GS(th)}$, is the

- A. maximum allowable gate-source voltage before breakdown.
- B. gate-source voltage that produces a leveling off of I_D .
- C. minimum gate-source voltage that makes drain current flow.
- D. None of the above.

40. A FET is also called _____ transistor.

- A. bipolar
- B. unipolar
- C. unijunction
- D. None of the above

41. When a JFET is operating in the ohmic region,

- A. I_D is independent of V_{DS} .
- B. I_D is independent of V_{BS} .
- C. I_D increases in direct proportion to V_{DS} .
- D. the drain acts like a current source.

42. The transconductance g_m of JFET is equal to

- A. $-2 \cdot \frac{I_{DSS}}{V_p}$
- B. $-2 \cdot \frac{I_{DSS}}{V_p} \left(1 - \frac{V_{GS}}{V_p}\right)$

C. $\frac{2}{|V_p|} \sqrt{I_{DSS} I_D}$

D. $\frac{I_{DSS}}{V_p} \left(1 - \frac{V_{GS}}{V_p}\right)$

43. The gate of a FET is _____ biased.

- A. forward
- B. reverse
- C. forward as well as reverse
- D. not

44. Which of the following JFET amplifiers has a high Z_{in} , a low Z_{out} , and a voltage gain less than one?

- A. The common-gate amplifier.
- B. The common-source amplifier.
- C. The source follower.
- D. Both A and B.

45. A FET is similar in operation to _____ valve.

- A. triode
- B. pentode
- C. tetrode
- D. diode

46. A JFET is a

- A. normally ON device.
- B. normally OFF device.
- C. bipolar device.
- D. current-controlled device.

47. The drain current I_D in a FET is given by _____.

A. $I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p}\right)^2$

B. $I_D = I_{DSS} \left(1 + \frac{V_{GS}}{V_p}\right)^2$

C. $I_D = I_{DSS} \left(1 - \frac{V_p}{V_{GS}}\right)^2$

D. $I_D = I_{DSS} \left(1 + \frac{V_p}{V_{GS}}\right)^2$

48. Given $I_D = 4 \text{ mA}$ at $V_{GS} = -2 \text{ V}$, determine I_{DSS} if $V_p = -5 \text{ V}$.

- A. 8.11 mA
- C. 9.11 mA

- B. 11.11 mA
- D. 10.11 mA

49. A FET is a _____ driven device.

- A. voltage
- B. current
- C. both voltage and current
- D. None of the above

50. The pinch-off voltage of a FET is about _____.

- A. 15 V
- B. 25 V
- C. 0.6 V
- D. 5 V

END